Numerical Computing II

Assignment IV

Lagrange interpolation

1. Given the data:

x	0	1	2
f(x)	1	3	7

1. Construct the Lagrange interpolating polynomial P■(x).
2. Write down each Lagrange basis polynomial L■(x), L■(x), L■(x).
3. Verify that L■(x■) = 1, if i = j; 0, if i ≠ j.
4. Use the polynomial to estimate f(1.5).
5. Compare your result with the exact value if $f(x) = x^2 + x + 1$.

2. Interpolation from experimental data:

The following data were obtained experimentally:

x	1	2	3	4
f(x)	2.0	4.5	8.0	13.5

1. Construct the interpolating polynomial P■(x).
2. Estimate f(2.5).
3. Estimate f(3.5).
4. Comment on whether the interpolated values appear reasonable based on the given data.

3. Non-uniformly spaced data:

x	0	0.5	1.5	3
f(x)	1.000	1.125	1.875	4.000

1. Construct the Lagrange interpolating polynomial.
2. Estimate f(1).
3. Estimate f(2).
4. Explain why Lagrange interpolation is particularly convenient for this data set compared with methods that require equally spaced points.

4. The following values of a function are known:

x	2	3	5	7
f(x)	5	10	26	50

Using Lagrange interpolation:

1. Construct P■(x).
2. Calculate f(4), f(6), f(8).
3. Which of the three estimates—f(4), f(6), f(8)—do you expect to be the most reliable?

5. Consider the data:

x	1	2	3	4
f(x)	1	4	9	16

A student says: “Since four data points are given, the interpolating polynomial must always be a cubic polynomial.”

1. Is the statement mathematically correct?
2. Construct the Lagrange interpolating polynomial.
3. What is the degree of the resulting polynomial?
4. Explain why the polynomial obtained can have degree less than $n-1$ even when n data points are given.